3DCP-Agent: an LLM agent for planning and evaluating 3D concrete printing through natural-language access to literature-trained prediction, inverse mix design and buildability assessment

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Abstract

Planning a 3D concrete printing (3DCP) job means answering, in sequence, questions that today live in different tools and different experts: what fresh and hardened properties a candidate mix will have, which published mixes have reached a target, whether a given material will stand under a given nozzle, geometry and schedule, and which mix would. This paper presents 3DCP-Agent, in which a large language model (LLM) is the interface and the planner, while every number comes from validated computational tools: 36 property models trained on a literature database of 3,274 papers (29,626 mixes, 236,199 measurements, 25,721 digitised curves) with paper-grouped cross-validation and conformal 80 % intervals; an inverse mix-design engine; a physics-based buildability calculator (Roussel plastic collapse with linear structuration, Suiker free-wall buckling, open time) calibrated on 7,420 published print runs; literature retrieval; and read-only SQL over both databases. The 16 tools are defined once in a registry and projected to the Anthropic, OpenAI and Model Context Protocol (MCP) schemas, so the same agent runs on API keys or on a user's Claude or ChatGPT subscription, in a chat web application, a terminal, or any MCP client. Guardrails in the system prompt forbid numbers that did not come from a tool, require intervals and units, and require stated defaults. Case studies show end-to-end print planning in conversation: per-cubic-metre batch sheets are converted to mix

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specifications, buildability verdicts come with layer cycle-time windows and alternatives, and a print job is turned into ranked mix candidates with published analogues. A benchmark of 33 natural-language tasks measures tool selection, parameter extraction against expected values, ground-truth answers for database questions, refusal of unsupported requests, resistance to write attempts, and numeric fidelity (the fraction of numbers in an answer traceable to a tool result). The agent and the evaluation harness are released with the package.

Keywords: 3D concrete printing, large language model, agent, tool use, model context protocol, mix design, buildability, literature database

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1. Introduction

Extrusion-based 3D concrete printing (3DCP) couples material design to process design more tightly than any conventional casting method. A printable mortar must be pumpable and extrudable at the nozzle, must carry the layers placed on top of it before it sets, and must still deliver strength, shrinkage and durability in the hardened state [1, 2]. The decisions a practitioner has to make are therefore distributed over several questions: what fresh and hardened properties a candidate mix is likely to have; which published mixes have already reached a given combination of targets; whether a given material will stand under a given nozzle, geometry and layer cycle time; and, in the inverse direction, which mix and which schedule a given object requires. Each of these questions has, by now, a quantitative answer in the literature: data-driven property models [3, 4, 5, 6], analytical buildability criteria [2, 7, 8, 9, 10] and large collections of published mix designs. In practice, however, the answers sit in separate scripts, spreadsheets and papers, each with its own input format and its own vocabulary, so the questions are rarely asked together and almost never by the people who print.

Large language models (LLMs) have changed what an interface to such tools can be. An LLM can read a mix description written the way a laboratory writes it ("500 kg cement, 150 kg fly ash, 1300 kg sand, 227 kg water per cubic metre"), convert it into the structured input a model expects, decide which computation answers the question, call it, and explain the result. Tool use (function calling) and the Model Context Protocol (MCP) [11, 12, 13, 14] make this orchestration reliable enough for engineering use, provided that the numbers themselves are produced by validated tools and not by the language

model [15]. Construction research has started to adopt this pattern for building information modelling, robotics, pavement management and structural health monitoring [16, 17, 18, 19, 20, 21], but not yet for the material-and-process planning problem of 3DCP, where the relevant tools are prediction models and physics calculators rather than databases and CAD.

This paper presents such an agent, which we call 3DCP-Agent. It is built on an open package that we developed for cement paste and mortar property prediction and inverse mix design from a validated literature database, and on a buildability layer calibrated on published print runs. The contribution of the present paper is the agent layer and its evaluation:

1. **A tool registry that makes the computational layer callable by any LLM.** 16 tools (prediction, rheology curves, literature retrieval, inverse design, buildability and schedule, job-to-mix design, schema inspection and read-only SQL, report and memory) are defined once with a JSON schema and projected to the Anthropic, OpenAI and MCP schemas. The same agent therefore runs on an API key, on a Claude subscription through the Claude Agent SDK, or on a ChatGPT subscription through the Codex CLI, and can be attached as an MCP server to third-party clients.
2. **A print-planning conversation as a closed loop.** A print job described in words (object, height, nozzle, speed, open time) is checked against a measured or predicted material with Roussel/Perrot/Suiker criteria and the empirical collapse statistics of 7,420 published runs; when it fails, the agent reports the layer cycle-time window and the required static yield stress or structuration rate; and the job can be turned into a design specification whose candidates are re-assessed against the job.
3. **Guardrails and a measurable notion of faithfulness.** The system prompt forbids any quantitative statement that did not come from a tool result in the same conversation, requires predictions as q_{50} $[q_{10}, q_{90}]$ with units, and requires every default the agent filled in to be stated. We operationalise this as *numeric fidelity*, the fraction of numbers in an answer that can be traced to the prompt, a tool input or a tool result, and measure it on every task.
4. **A reproducible benchmark.** 33 tasks in ten groups, each with machine-checkable expectations (tools called, parameters extracted, ground-truth answers for database questions, refusal of an unsupported property, rejection of a write attempt, persistence of a remembered

preference), are run from a fresh session on each backend and scored without human judgement; the harness and the tasks are part of the released package.

The remainder of the paper reviews related work (Section 2), describes the data and computational layer, the agent and the implementation (Section 3), presents case studies and the benchmark (Section 4), discusses limitations (Section 5) and concludes (Section 6).

2. Related work

2.1. Tool-using LLM agents and the Model Context Protocol

Toolformer showed that a language model can learn when to call external tools and how to use their results [11]; ReAct interleaves reasoning traces with actions and observations so that a model plans, acts and revises in one loop [12]. Function calling, the production form of these ideas, lets a model emit a structured call against a JSON schema and receive the result as a message. In chemistry, ChemCrow demonstrated that an LLM equipped with expert tools solves synthesis and design tasks that the bare model cannot, and that the value lies in the tools, not in the model's recollection of facts [15]. Retrieval-augmented generation addresses the complementary problem of grounding text answers in documents [22]. The Model Context Protocol, released in November 2024, standardises how a host application discovers and invokes tools, resources and prompts offered by a server, so that one server can be used by many clients and one client can combine many servers [13, 14]. The present work uses all three mechanisms: function calling against a registry, MCP for clients we do not control, and a lightweight retrieval of published analogue mixes.

2.2. LLMs and agents in construction and civil engineering

Table 1 summarises recent construction-domain systems. Text2BIM generates and edits building models from natural language through multi-agent code generation [16]; Dong et al. combine an LLM with BIM tooling for AI-assisted modelling [17]; Gao et al. query and manipulate IFC models through an agent [23]; Chen and Bao orchestrate robotic construction with several agents [18]; Ren et al. apply agentic control in engineering informatics [24]; Zeng et al. use agents for pavement management [19]; Liang et al. address structural engineering tasks [25]. Closest to our setting are two MCP-based systems: Avila et al. connect GPT-4 to OpenSeesPy for structural analysis

Table 1: Recent LLM systems in construction and materials, and the present work. KB: knowledge base; MCP: model context protocol.

Reference	Domain	LLM role	Tools	MCP	KB	Evaluation
Du et al. [16]	BIM authoring	multi-agent code generation	yes	no	no	case + metrics
Dong et al. [17]	BIM modelling	assistant	yes	no	no	case
Gao et al. [23]	IFC querying	agent	yes	no	no	case + metrics
Chen, Bao [18]	robotic construction	multi-agent orchestration	yes	no	no	case
Zeng et al. [19]	pavement management	multi-agent	yes	no	yes	case
Avila et al. [21]	structural analysis	GPT-4 + OpenSeesPy	yes	yes	no	case
Liu et al. [20]	dam SHM	orchestrator of 4 agents	yes	yes	RAG	case, timing
Li et al. [30]	underground concrete	Q&A	no	no	no	expert rating
Völker et al. [31]	concrete mix design	direct generation	no	no	no	lab tests (preprint)
This work	3DCP planning	planner + interface over validated tools	16	yes	2 DBs	cases + 33-task benchmark

[21], and Liu et al. build a multi-agent system for dam structural health monitoring in which the LLM orchestrates deep-learning behaviour models, an anomaly detector, a data-management agent and a RAG knowledge base through MCP [20]. In the materials domain, LLMs have mostly been used for information extraction from the literature [26, 27, 28, 29] or evaluated as question-answering assistants [30]; direct LLM design of concrete has been proposed in a preprint [31] and LLMs have been combined with vision for process monitoring [32] and with ML strength models [33]. None of these systems gives an LLM tool access to property prediction, inverse design and buildability physics for 3DCP, and none reports a task-level benchmark of the agent layer itself; the dam-monitoring and structural-analysis papers evaluate their systems through case studies.

2.3. Data-driven mix design and buildability in 3DCP

Machine learning for 3DCP has mostly targeted single properties: slump flow and printability [3], rheology from small datasets with ensembles [4], interlayer tensile strength [5], bond and reinforcement design [34], and generative design of printable mixes [35]; Monkman et al. review data-driven mix design and note the shortage of large, consistently extracted datasets

[6]. Buildability has an analytical core: the bottom layer of a stack yields when its vertical stress exceeds $\sqrt{3}$ times the static yield stress, which itself grows during the print [2, 7, 36]; slender walls fail earlier by elastic buckling [8, 9]; Kruger et al. formulated a lower-bound model and characterised thixotropy for these criteria [10, 37]. Our computational layer implements these criteria and calibrates them on published collapse tests; what is new here is that they are reachable through conversation together with the prediction models that supply their inputs.

3. Methodology

3.1. Overview

Figure 1 shows the five layers of the system. Users reach the agent through a chat web application, a terminal, or an external MCP client. The agent core builds a system prompt with live context, runs the tool loop, and keeps session memory. All tools are defined once in a registry and projected to the schema each LLM backend expects. The computational layer is the existing prediction and design package: property models, inverse design, buildability physics, literature retrieval and guarded SQL. The data layer holds the literature database and the print-process database. The LLM never computes a property; it decides which tool to call with which arguments, and it explains the results.

Figure 2 shows how one print-planning conversation maps onto tool calls: the LLM parses the job and the material, calls the buildability tool, explains the verdict, turns the job into a design specification on request, and saves a report.

3.2. Data layer

Literature database. The literature database was built in our previous work with an LLM-assisted extraction pipeline in which every paper is turned into a structured record (papers, materials with chemistry and physical properties, mixes and components, tests and protocols, measurements with reported and canonical units, digitised figure curves, unmapped terms and quality exceptions) and every record must pass a frozen machine validator before it enters the database; a sample of papers is re-verified against the source by independent agents and the error rate is reported. The pipeline follows the structured-extraction approach of [26, 27]. At the time of writing the database holds 3,274 papers (3,117 included after full-text screening, 1,176

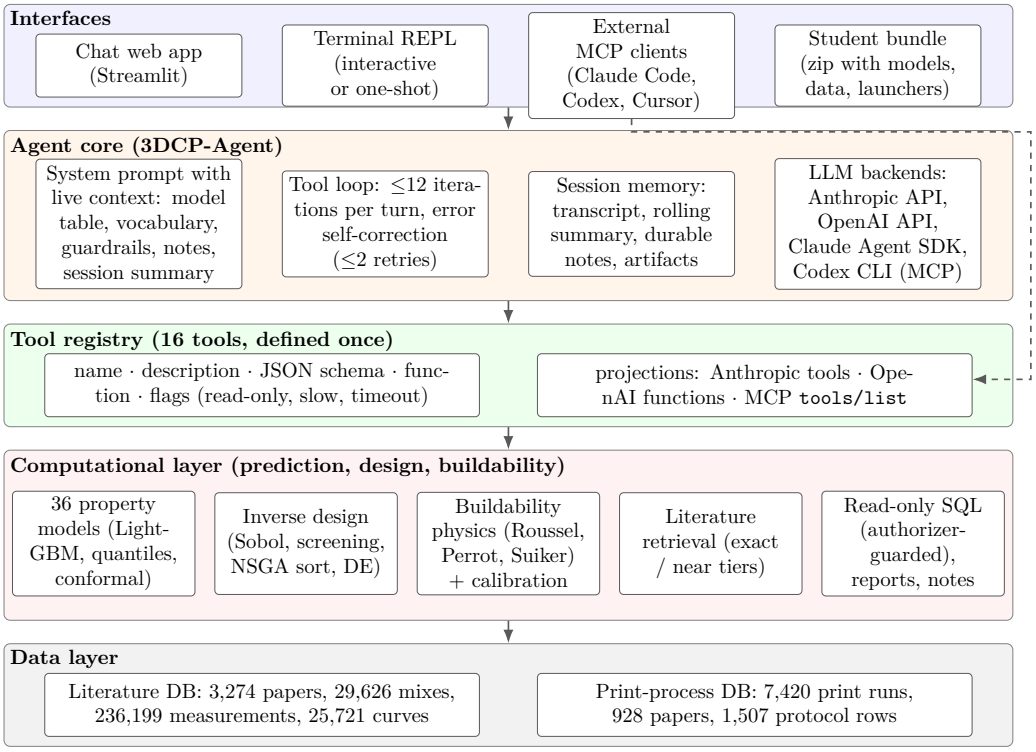


Figure 1: System architecture. The LLM (agent core) plans and explains; every quantitative result comes from the computational layer through a tool whose schema is defined once and projected to the Anthropic, OpenAI and MCP formats.

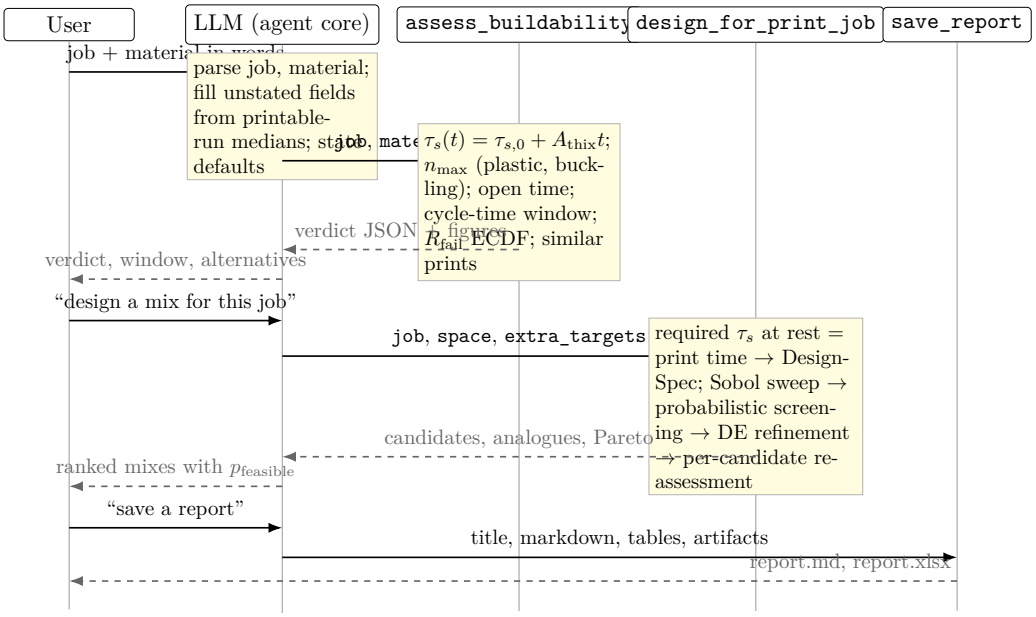


Figure 2: A print-planning conversation as a sequence of tool calls. Solid arrows are structured requests, dashed arrows are results; the notes summarise what each tool computes.

of them 3DCP studies), 29,626 mixes, 236,199 measurement rows of which 81,030 carry a tabulated value, and 25,721 digitised curves with 155,211 points. Units are harmonised per quantity after merging, and the basis of every ratio (per gram of binder, of paste, of cement) is recorded explicitly.

Print-process database. For the buildability layer, 928 3DCP papers were reread for printing-process variables and outcomes: printer and nozzle, layer height and width, print speed, layer cycle time, number of layers, height reached, start time after mixing, open time, outcome and failure mode, and the authors' own verdict as a printability label with its basis and confidence. The result is 7,420 print runs (5,586 printable, 932 borderline, 902 non-printable), 941 collapses, 841 stack-to-failure tests, and 1,507 rheology-protocol rows that record how each static yield stress was measured (device, rest time). Runs are keyed to the mixes of the literature database so that fresh properties and print outcomes can be joined.

3.3. Computational layer

Property models. A mix is represented by 138 features: binder composition by material class, water-to-binder and sand-to-binder ratios, admixture, fi-

bre and nano dosages, oxide chemistry and fineness of the materials (class medians when a paper does not report them), and test conditions (age, specimen comparability group, curing, rest time, 3DCP flag). 28,001 of the 29,626 mixes could be normalised to this representation. For each of 19 target quantities a bundle is trained: a LightGBM point model [38], q_{10} and q_{90} quantile models whose width is rescaled by split-conformal calibration [39], and a paper-bootstrap ensemble for an epistemic spread. A second, 3DCP-specific variant is trained for targets with enough 3DCP rows and is selected automatically when a query is flagged as 3DCP and the variant's grouped-CV quality is higher, giving 36 models in total. All quality metrics are computed with GroupKFold by paper, so no mix of a test paper is seen in training [40]; Figure 3 shows the rank correlation and training size per target. Models flagged *weak* (few papers, poor interval coverage or low rank correlation; most rheology and flow targets) are reported to the LLM in the system prompt, which is instructed to present them as order-of-magnitude guidance and to lean on published analogues.

Inverse design. A design specification lists targets (at least, at most, range, goal; with units, conditions and a minimum probability of satisfaction), a search space (system type, w/b and s/b ranges, allowed binder classes with bounds and required flags, admixture ranges, curing) and objectives (clinker fraction, embodied CO_2, cost, or per-MPa variants). The engine draws a Sobol sample of the space, predicts every target with its interval, screens candidates by the product of per-target satisfaction probabilities, performs a non-dominated sort on objectives and feasibility, selects a diverse shortlist and refines it by differential evolution over the forward models. Every candidate is returned with its applicability-domain distance and with the closest published mixes as analogues. Curve-derived targets (shear stress at a given shear rate from the Bingham flow curve; static yield stress after a given rest time from the structuration curve) are available as targets, which is what links inverse design to buildability.

Buildability and schedule. A print job is described by object type (wall, hollow cylinder, column, other), target height H, footprint (wall length or cylinder diameter), number of filaments across the wall thickness, nozzle, layer height h and width, print speed, layer cycle time t_c, start time after mixing, open time and a load safety factor s_f. Unstated fields are filled from the medians of printable published runs ($h = 0.5\,d_{\text{nozzle}}$, $w = 1.2\,d_{\text{nozzle}}$, 40 mm/s, t_c

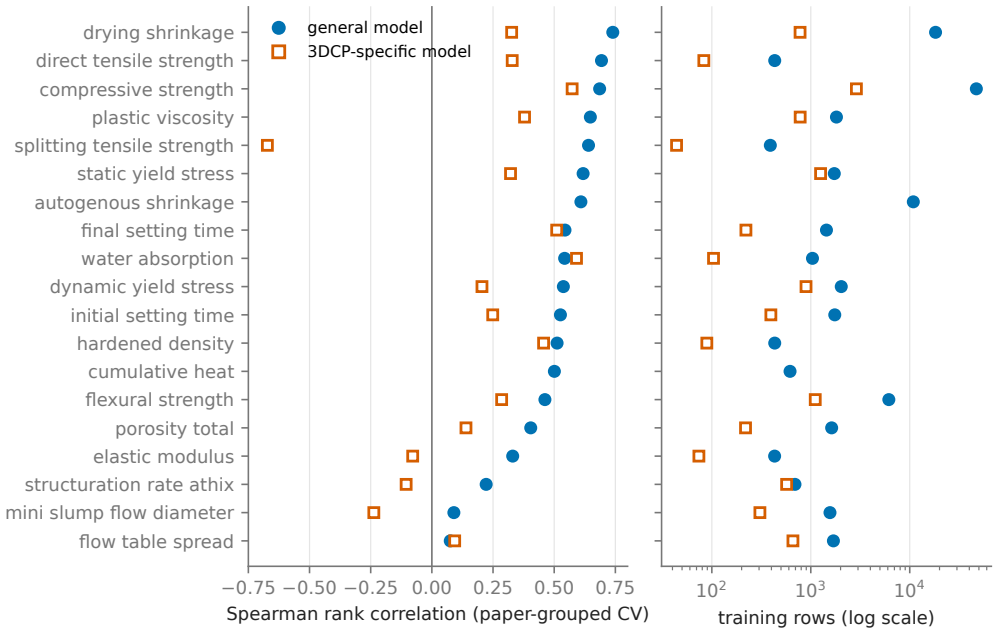


Figure 3: Paper-grouped cross-validation quality (Spearman rank correlation between predicted and measured values, left) and training size (right) of the general and 3DCP-specific property models. Low-ranked targets are flagged weak in the system prompt.

= path length / speed) and the agent is required to say so. The fresh material is given either as measured values ($\tau_{s,0}$, A_{thix}, density, E/τ_s) or as a mix, in which case the property models supply them. With linear structuration $\tau_s(t) = \tau_{s,0} + A_{\text{thix}}\, t$ [7, 2], the bottom layer of an n-layer stack built at one layer per t_c yields when $\rho g n h > \sqrt{3}\,\tau_s(n t_c)$, so that

$$n_{\max}^{\text{plastic}} = \frac{\sqrt{3}\,\tau_{s,0}}{\rho g h / s_f - \sqrt{3} A_{\text{thix}} t_c}, \qquad \text{unlimited if } \sqrt{3} A_{\text{thix}} t_c \geq \rho g h / s_f. \tag{1}$$

For a free straight wall clamped at its base the critical height follows Suiker's self-weight buckling solution with the green modulus $E(t) = (E/\tau)\,\tau_s(t)$ [8]:

$$l_{\text{cr}} = \left(\frac{7.8373\, E(t)\, t_w^2}{12\, \rho g} \right)^{1/3}, \tag{2}$$

with t_w the wall thickness (filaments $\times$ layer width); closed sections are reported as not buckling-critical. The open-time check requires start time

plus print duration to stay within the open time. The verdict (printable, borderline, non-printable) names the governing mode and reports $n_{\max}$ with and without the safety factor; the *schedule* sweeps t_c and print speed and returns the window between the minimum cycle time required for stability and the maximum allowed by the open time, or, when the window is closed, the $\tau_{s,0}$ and A_{thix} that would open it.

The criteria are calibrated on the print-process database. For the 146 stack-to-failure collapses (37 papers) whose mix has a tabulated static yield stress, the ratio $R_{\text{fail}} = \rho g H_{\text{collapse}}/(\sqrt{3}\,\tau_s)$ has median 1.39 with interquartile range 0.64–2.87 (Figure 4); $R = 1$ is the static criterion with $\tau_{s,0}$ alone, and the excess is the structuration accrued during the print. The empirical distribution function of R_{fail} turns the static ratio of a job into an empirical collapse probability that is reported next to the physical verdict. The tabulated static yield stress alone separates collapsed from successful runs with an area under the ROC curve of only 0.59 over 611 runs, because published values mix rest protocols and because structuration, not $\tau_{s,0}$, carries tall stacks; this is why the physics route with structuration is the primary verdict. Extrudability windows (10th–90th percentiles of printable runs: τ_s 273–4920 Pa, plastic viscosity 2.7–57 Pa·s, flow table spread 126–197 mm) are used as caution flags, not hard limits. Finally, `design_for_print_job` converts a job into a design specification by adding the derived target "static yield stress after a rest time equal to the print duration $\geq s_f\,\rho g H/\sqrt{3}$", an A_{thix}-maximising steering objective and the extrudability window of the nozzle class, runs the inverse design, and re-assesses every candidate against the job.

Literature retrieval and SQL. Literature retrieval ranks published mixes against the same target list used for design: tier *exact* contains mixes whose measured values satisfy every target; tier *near* contains the closest misses. For arbitrary questions, a schema-inspection tool returns tables, columns and sample rows of either database, and a SQL tool executes a single SELECT statement under two independent guards (a keyword filter that rejects anything but SELECT/WITH, and an SQLite authorizer that denies every write and schema operation), with a 20 s timeout and a 200-row cap; results above 20 rows are returned to the model as a digest and saved in full as CSV.

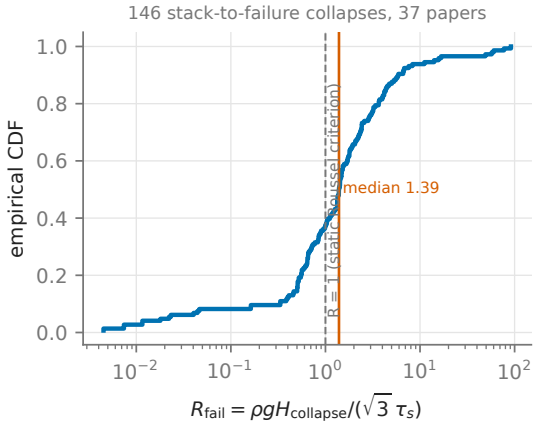


Figure 4: Empirical distribution of the collapse ratio R_{fail} in published stack-to-failure tests. The agent reports the empirical collapse probability of a job's static ratio next to the physical verdict.

3.4. Tool registry and schema projection

Each tool is registered once with a name, a description, a JSON schema for its arguments, the Python function, and flags (read-only, slow, timeout). The registry is projected to the Anthropic tool format, the OpenAI function format and the MCP `tools/list` response, so the three are consistent by construction (a test asserts their equality). Table 2 lists the 16 tools. Every tool returns a `ToolResult` with a JSON data block, a one-line summary and a list of artifacts (tables, images, markdown, xlsx) written to the session's artifact directory; the model sees at most 20 rows of a table and a truncated JSON, the interface renders the artifacts with download buttons. Natural-language dictionaries are normalised by the same rules the former form interface used: binder fractions are renormalised to one with a warning, admixtures are percentages of binder by product mass, curing can be a preset name or a word (`water`, `moist`, `sealed`, `ambient`, `steam`), and an unknown material class raises an error that names the closest valid classes so that the model can correct itself.

3.5. Agent core

System prompt with live context. The system prompt is rebuilt for every turn from the live state of the package: the role, the data sizes, the model table

Table 2: The tool registry. "slow" tools take 30–150 s; all tools are read-only with respect to the databases.

tool	kind	purpose	inputs
`retrieve_literature`	read	published mixes near target values	`targets`, `system_type`, `prefer_3dcp`, `n`
`design_mix`	slow	inverse mix design (Sobol → screening → DE)	`targets`, `space`, `objectives`, `budget`, `top_n`, `n_literature`, `name`
`build_mix_spec`	read	validate a mix description	`mix`
`predict_properties`	read	properties with 80 % intervals	`mix`, `targets`
`flow_curve`	read	Bingham flow curve + measured analogues	`mix`, `max_rate_1s`, `n_analogues`, `log_y`
`structuration_curve`	read	$\tau_s(t)$ + measured analogues	`mix`, `max_rest_min`, `n_analogues`
`assess_buildability`	read	verdict, schedule window, similar prints	`job`, `material`, `mix`, `rho_kg_m3`, `E_over_tau`, `n_similar`
`print_schedule`	read	layer cycle-time / speed window	`job`, `material`, `mix`
`similar_prints`	read	nearest published print runs	`job`, `material`, `k`
`design_for_print_job`	slow	job → required rheology → mixes → re-assessment	`job`, `space`, `extra_targets`, `budget`, `rho_kg_m3`
`remember`	write	durable note injected into future prompts	`note`, `kind`
`list_models`	read	predictable targets and model quality	`variant`
`describe_vocabulary`	read	material classes, specimen groups, curing presets	`family`, `query`
`save_report`	write	markdown / xlsx report of a session	`title`, `markdown`, `tables`, `include_artifacts`, `format`
`describe_schema`	read	tables, columns, samples of the databases	`db`, `table`
`sql_query`	read	read-only SELECT (authorizer-guarded)	`sql`, `db`, `max_rows`

with training size, grouped-CV quality and weak flags, the material-class vocabulary, specimen groups and curing presets, a mix-dictionary example, the guardrails, the user's durable notes and the rolling summary of the session. Because the model table is generated from the trained artifacts, retraining the models changes what the agent promises without any prompt edit.

Guardrails. Nine rules are stated in the prompt. The ones that matter for engineering use are: (1) never invent numbers — every quantitative statement must come from a tool result in this conversation, and predictions are written as $q_{50}\,[q_{10}, q_{90}]$ with units; (2) targets flagged weak are presented as order-of-magnitude guidance with published analogues as primary evidence, and extrapolation beyond the training range (reported by the tool as range violations) is stated; (3) defaults filled in for values the user did not give are named; (4) database questions go through schema inspection before SQL, with explicit distinction between paper counts and mix counts; (5) slow tools run immediately only when targets and search space are given, otherwise one clarifying question is asked, once; (6) a non-printable verdict is accompanied

by actionable alternatives (cycle-time window, layer height, required $\tau_{s,0}$ or A_{thix}, accelerator, VMA, lower w/b); (8) a tool error is explained in one line and the input is corrected and retried at most twice; (9) the models are correlations in published data and do not replace experiments.

Tool loop and memory. A turn runs until the model stops calling tools or twelve iterations are reached. Transcripts, metadata and a rolling summary are stored per session; when a transcript exceeds 40 000 characters the older part is summarised by a cheap model call and replaced by the summary in the prompt. Notes saved with the `remember` tool are appended to a durable file that enters the system prompt of every later session, which is how a laboratory's conventions (default specimen, preferred units) persist. Sessions can be resumed from the interface or the command line.

3.6. LLM backends and interfaces

The provider interface is a single generator per turn that yields text deltas, tool calls, tool results and a final usage record. Four providers implement it: the Anthropic Messages API and the OpenAI Chat Completions API with native function calling (tools executed in-process); the Claude Agent SDK, which runs the bundled Claude Code CLI under the machine's Claude subscription and serves the registry as an in-process MCP server; and the OpenAI Codex CLI, which runs under a ChatGPT subscription and reaches the tools through a stdio MCP server started as a child process. The provider is chosen in the order API key, Claude login, ChatGPT login, or set explicitly. Two details of the Codex path were necessary in practice: recent versions of the CLI refuse MCP tool calls in non-interactive mode unless the server pre-approves its tools, and the CLI blocks on an open standard-input pipe, so it is started with a closed stdin. A fifth, scripted provider supports tests without an LLM. The web application (Streamlit) is a chat with inline tool status, rendered tables and figures, download buttons, backend and session selection and example prompts; a terminal REPL and a one-shot mode share the same engine; the MCP server can be attached to Claude Code, Codex or Cursor directly. Prompts, labels and the system prompt are available in Korean and English. For distribution to students the package is zipped with the trained models, both databases, launcher scripts and a setup guide; the only external requirement is Python and, for the ChatGPT path, Node.js for the Codex CLI.

4. Results

4.1. Case studies

All transcripts below are verbatim outputs of the released system on the ChatGPT-subscription backend (Codex CLI, model gpt-6-astra) recorded by the benchmark harness; tool calls are shown with their arguments and timing. The only edit is that the package's internal name, which the model sometimes repeats from the tool descriptions, is replaced by the system name. The figures shown with the cases were produced by re-executing the recorded tool calls with the same arguments in the English interface setting; the tools are deterministic, so they are the figures the agent offered for download, with English labels.

From a batch sheet to a prediction. Figure 5 shows the most common laboratory input, a per-cubic-metre batch sheet. The agent derives the ratios the model needs ($w/b = 227/650 = 0.349$, $s/b = 2.0$, binder fractions 0.77/0.23, PCE 0.5 % of binder), calls `predict_properties` once for both targets, and reports each prediction with its 80 % interval and the defaults it assumed (28 d, 50 mm cube, moist curing). The width of the shrinkage interval, more than a factor of two, is a property of the literature pool that the model is instructed to pass on rather than hide.

User My mix per cubic metre is: cement 500 kg, fly ash (class F) 150 kg, river sand 1300 kg, water 227 kg, PCE 3.25 kg. Predict 28-day compressive strength and drying shrinkage.

tool `describe_vocabulary`, 0.0 s


```
{}
```


tool `build_mix_spec`, 0.0 s


```
{"mix": {"system_type": "mortar", "w_b": 0.34923076923076923, "s_b": 2, "binder": {"
    portland_cement": 0.7692307692307693, "fly_ash_class_F": 0.23076923076923078}, "
    sand_class": "natural_sand", "admixtures_pct": {"superplasticiser_pce": 0.5}, "
    conditions": {"ag ...}
```


tool `predict_properties`, 2.3 s


```
{"mix": {"system_type": "mortar", "w_b": 0.34923076923076923, "s_b": 2, "binder": {"
    portland_cement": 0.7692307692307693, "fly_ash_class_F": 0.23076923076923078}, "
    sand_class": "natural_sand", "admixtures_pct": {"superplasticiser_pce": 0.5}, "
    conditions": {"ag ...}
```

Agent I'll check the model's input conventions and predict both properties, stating any assumptions needed for curing, specimen type, and PCE dosage.

Property at 28 days	q50 [q10, q90]
Compressive strength	**49.7 [31.2, 75.0] MPa**
Drying shrinkage	**854 [237, 1,577] microstrain**

Brackets show the central 80% prediction interval. Neither prediction exceeded the model's training ranges.

Assumptions: ordinary Portland cement, river sand classified as `natural_sand`, conventional mortar, `comp_cube50` specimens, and moist curing at 20 degC / 95% RH. Your proportions give **w/b = 0.349**, **sand/binder = 2.00**, and **PCE = 0.50% of binder by product mass**; no additional water correction was made for the PCE.

Drying shrinkage was evaluated with the age input set to 28 days; drying exposure and specimen details were not specified, so it is not a test-protocol-specific estimate. Both predictions are literature-based correlations and need experimental verification.

Files: prediction CSV, prediction figure, mix specification.

Figure 5: Case: per-cubic-metre batch sheet → structured mix → prediction with intervals and stated defaults.

Rheology curves with published analogues. Figure 6 asks for a Bingham flow curve and a structuration curve of a 3DCP mortar. The agent calls the two curve tools with the requested ranges (100 s^{-1}, 45 min); each returns a figure in which the predicted curve and its band are overlaid with measured curves of the closest published mixes, and a CSV of the points. Figure 7 shows the two figures as produced.

User For a 3DCP mortar with w/b 0.35, s/b 1.5, 90 % OPC + 10 % silica fume and 0.6 % PCE, plot the Bingham flow curve up to 100 1/s and the structuration curve over 45 minutes of rest, and compare both with measured literature curves.

tool `describe_vocabulary`, 0.0 s

```
{}
```

tool `describe_vocabulary`, 0.0 s

```
{"query": "superplasticiser_pce"}
```

tool `build_mix_spec`, 0.0 s


```
{"mix": {"name": "OPC90_SF10_PCE06", "system_type": "mortar", "w_b": 0.35, "s_b": 1.5,
      "binder": {"portland_cement": 0.9, "silica_fume": 0.1}, "admixtures_pct": {"
     superplasticiser_pce": 0.6}, "conditions": {"is_3dcp": true, "rest_time_s": 0}}}
```


tool `flow_curve`, 4.4 s


```
{"mix": {"name": "OPC90_SF10_PCE06", "system_type": "mortar", "w_b": 0.35, "s_b": 1.5,
      "binder": {"portland_cement": 0.9, "silica_fume": 0.1}, "admixtures_pct": {"
     superplasticiser_pce": 0.6}, "conditions": {"is_3dcp": true, "rest_time_s": 0}},
     "max_rate_1s": 1 ...}
```


tool `structuration_curve`, 0.8 s


```
{"mix": {"name": "OPC90_SF10_PCE06", "system_type": "mortar", "w_b": 0.35, "s_b": 1.5,
      "binder": {"portland_cement": 0.9, "silica_fume": 0.1}, "admixtures_pct": {"
     superplasticiser_pce": 0.6}, "conditions": {"is_3dcp": true, "rest_time_s": 0}},
     "max_rest_min":  ...}
```


tool `describe_schema`, 0.7 s


```
{"db": "master"}
```


tool `describe_schema`, 0.0 s


```
{"table": "curves"}
```


… 3 further tool calls omitted

Agent I'll check the mix conventions, predict the flow and structuration curves, and retrieve measured literature curves for comparison.
Created both plots with literature overlays and q10-q90 uncertainty bands:

- **Bingham flow:** Plot . CSV
- **Structuration:** Plot . CSV

Predicted quantity	q50 [q10, q90]
Dynamic yield stress	234 [16, 843] Pa
Plastic viscosity	6.43 [0.84, 33.13] Pa.s
Shear stress at 100 s^-^1	1,023 [250, 3,901] Pa
Initial static yield stress	271 [22, 9,125] Pa
Structuration rate	0.697 [0.023, 13.669] Pa/s
Static yield stress after 45 min	4,003 [307, 61,123] Pa

The curves use \(\tau=\tau_0+\mu\dot\gamma\) and \(\tau_s=\tau_{s0}+A_{\mathrm{thix}}t\). Pointwise quantiles come from sampled parameter distributions; they need not equal curves constructed from parameter medians.
Literature comparison. The flow overlays include published silica-fume mortars at w/b 0.35 and s/b 1. Their stresses at 100 s^-^1 are 202.2 Pa for 6% silica fume and 309.17 Pa for 10%–below the predicted median. These overlays reconstruct the authors' fitted Bingham equations, rather than raw measured points. The NZ-0 overlay uses digitized measured flow curves.
The structuration overlays are digitized measurements for R20, R40 and R60. Their recorded stress ranges are respectively 2,080-3,650, 970-2,130 and 170-1,080 Pa over 10-60 minutes. **Their clock starts at water addition, not at the start of rest**, so they provide qualitative context rather than a matched validation.
The structuration-rate model is weak: treat its curve as **order-of-magnitude guidance**, with the published measurements as primary evidence. Experimental verification is needed.
Defaults: natural sand; PCE interpreted as product mass relative to binder; initial rest time zero. The tool also populated age 28 d, `comp_cube50`, and moist curing at 20 degC/95% RH; these are metadata defaults, not a fresh-rheology test protocol.

Figure 6: Case: flow curve and structuration curve of a 3DCP mortar with measured literature analogues.

Buildability verdict and schedule window. Figure 8 is the central planning question: a 600 mm wall, 1.5 m long, two filaments, 25 mm nozzle, 50 mm/s,

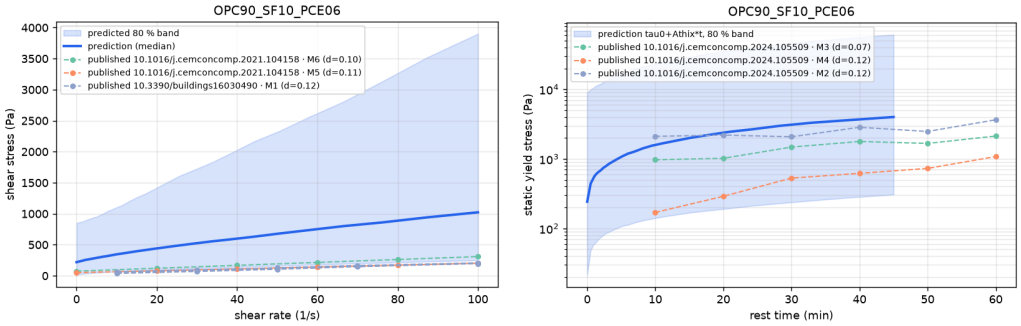


Figure 7: Figures returned by the `flow_curve` and `structuration_curve` tools for the conversation of Figure 6: predicted curve with 80 % band and the measured curves of the closest published mixes.

60 min open time, and a material with $\tau_{s,0} = 3$ kPa and $A_{\text{thix}} = 1$ Pa/s. The agent converts the units, calls `assess_buildability`, and reports the verdict with the governing mode, the number of layers the material carries with and without the safety factor, the cycle-time window, the empirical collapse probability and the closest published prints. Figure 9 shows the structured request and the digest of the response exchanged between the LLM and the tool for this turn; Figure 10 the cycle-time sweep the tool produced.

User I want to print a 600 mm high, 1.5 m long wall with two filaments across the thickness, 25 mm nozzle, 50 mm/s, open time 60 min. The material has a static yield stress of 3 kPa and Athix of 1 Pa/s. Is it printable, and what is the layer cycle-time window?

tool `assess_buildability`, 2.1 s

```
{"job": {"object_type": "wall", "target_height_mm": 600, "footprint_mm": 1500, "
    wall_filaments": 2, "nozzle_d_mm": 25, "print_speed_mm_s": 50, "open_time_min":
    60}, "material": {"tau_s0_Pa": 3000, "athix_Pa_s": 1}}
```

Agent I'll check the wall's buildability and layer cycle-time window using the physical model, including collapse, buckling, and open time.

Not printable as an unsupported straight wall under the calculator's default assumptions. There is no feasible layer cycle-time window.

Check	Calculated result
Current cycle, including both filaments	60 s/layer
Total printing time	48 min
Minimum cycle for plastic stability, safety factor 1.5	160.5 s/layer
Maximum cycle allowed by open time	75.0 s/layer
Feasible window	**None: minimum exceeds maximum**

Buckling governs: the free-wall model estimates instability at approximately **212.5 mm**. Plastic collapse also prevents reaching the target; its estimated height limit without the safety factor is **423 mm**. Even without that factor, the plastic-stability minimum cycle is **86.2 s**, exceeding the open-time limit.

Assumed defaults: **12.5 mm layer height**, **29.61 mm filament width**, approximately **59 mm total thickness**, density **2,100 kg/m^3**, **E/tau = 25**, immediate printing after mixing, and no dwell or travel overhead.

To proceed:

- Add bracing, returns, or wall thickness to address buckling.
- At the current **60 s** cycle, the plastic-stability calculation requires either initial static yield stress **>=7.82 kPa** with Athix unchanged, or **Athix >=2.68 Pa/s** with initial yield stress unchanged. These thresholds **do not resolve buckling**.
- Accelerator, VMA, or lower w/b are possible material adjustments, subject to extrusion testing.

These are deterministic physical calculations, not prediction intervals. Confirm fresh stiffness, density, and stability experimentally.

Files: Assessment, cycle plot, cycle CSV, literature analogues.

Figure 8: Case: buildability verdict and layer cycle-time window for a wall from measured fresh properties.

From a print job to ranked mixes. Figure 11 turns the same job into a design task. The agent passes the job, the search space and the strength requirement to `design_for_print_job`; the tool derives the required static yield stress after the print duration, runs the inverse design and re-assesses each candidate against the job. For this job the honest answer is negative: among 4 096 sampled mixes none satisfies the rheology requirement with the models' wide intervals, the tool returns its best fallback candidate, and the agent re-assesses it, reports it explicitly as a laboratory trial mix and not as an approved printing mix, states the required static yield stress and structuration rate, and retrieves published mixes near the required rheology as evidence for the next trials. Figure 12 shows the three-turn version of the same workflow (assess, design, save a report) in which the job is carried in session memory between turns; the saved report lists the assumptions, the assessment, the unsuccessful search, the fallback mix and the literature analogue.

User Design a mix for this print job: 600 mm high, 1.5 m long wall, two filaments, 25 mm nozzle, 50 mm/s, open time 60 min. Space: OPC required 0.5-1.0, silica fume 0-0.15, limestone powder 0-0.3, w/b 0.28-0.42, s/b 0.8-1.8, PCE 0.1-1.2 %, cellulose VMA 0-0.4 %. The mix must also reach 30 MPa at 28 days. Run without asking.

tool `describe_vocabulary`, 0.0 s


```
{"family": "all"}
```


tool `design_for_print_job`, 42.0 s


```
{"job": {"object_type": "wall", "target_height_mm": 600, "footprint_mm": 1500, "
    wall_filaments": 2, "nozzle_d_mm": 25, "print_speed_mm_s": 50, "open_time_min":
    60}, "space": {"system_type": "mortar", "binder": {"portland_cement": {"lo": 0.5,
     "hi": 1, "required ...}
```


tool `retrieve_literature`, 0.6 s


```
{"system_type": "mortar", "prefer_3dcp": true, "n": 3, "targets": [{"quantity": "
    compressive_strength", "kind": "ge", "lo": 30, "unit": "MPa", "age_d": 28, "
    comparability_group": "comp_cube50"}]}
```


tool `predict_properties`, 1.0 s


```
{"mix": {"system_type": "mortar", "binder": {"portland_cement": 0.78, "
    limestone_powder": 0.22}, "w_b": 0.286, "s_b": 1.73, "admixtures_pct": {"
    superplasticiser_pce": 1.02}, "conditions": {"age_d": 28, "comparability_group":
    "comp_cube50", "curing": "moist", " ...}
```

tool `assess_buildability`, 0.6 s


```
{"job": {"object_type": "wall", "target_height_mm": 600, "footprint_mm": 1500, "
    wall_filaments": 2, "nozzle_d_mm": 25, "print_speed_mm_s": 50, "open_time_min":
    60}, "mix": {"system_type": "mortar", "binder": {"portland_cement": 0.78, "
    limestone_powder": 0.22}, ...}
```

Agent I'll run the print-job design within your specified mix limits, then check buildability and the strength target. I'll report any defaults used for unspecified printing or test conditions.
The search found no candidate meeting the original screening criteria. Its best buildability candidate fails both plastic-collapse and free-wall buckling checks, and the required delay between layers exceeds the open-time allowance. I'm checking that candidate's strength and the material improvements needed before presenting it as a trial mix.
No feasible mix was found for the specified job. The search evaluated 4,096 mixes; candidates returned after automatic screening relaxation still failed physical buildability checks.
The best-ranked buildability candidate, rounded and independently reassessed, is a **laboratory trial mix–not an approved printing mix**:

Component or ratio	Trial value
OPC, binder mass fraction	0.78
Limestone powder, binder mass fraction	0.22
Silica fume	Omitted
w/b	0.286
s/b	1.73
PCE, product mass/binder mass	1.02%
. . . 1 further rows in the transcript	

Predictions are q50 [q10, q90]:

Property	Prediction
Compressive strength, 28 days	**55.0 [34.2, 96.7] MPa**
Initial static yield stress	563 [40, 9,654] Pa
Structuration rate	0.778 [0.016, 14.605] Pa/s
Plastic viscosity	6.73 [0.99, 48.03] Pa.s

Strength is promising, but **plastic collapse governs**. The reassessment indicates collapse around layer 6 and free-wall buckling around layer 11, against 48 required layers. The minimum stability cycle is approximately **272 s**, while open time permits at most **75 s**: there is no feasible cycle-time window.
Defaults used: layer height **12.5 mm**, filament width **29.61 mm**, **60 s** cycle, no dwell, printing immediately after mixing, density **2,100 kg/m^3**, safety factor **1.5**, and stiffness/yield-stress ratio **25**. Strength assumes moist curing and `comp_cube50`.
To progress, experimentally target **Athix >=3.52 Pa/s** at the predicted initial yield stress, or **initial static yield stress >=8,464 Pa** at the predicted structuration rate. These are alternative plastic-collapse requirements; buckling still needs bracing, returns, or greater wall thickness. VMA adjustment within your range is worth testing; an accelerator would require expanding the search space.
[...]

Tool request (client → server)


```
{
 "tool": "assess_buildability",
 "arguments": {
  "job": {
   "object_type": "wall",
   "target_height_mm": 600,
   "footprint_mm": 1500,
   "wall_filaments": 2,
   "nozzle_d_mm": 25,
   "print_speed_mm_s": 50,
   "open_time_min": 60
  },
  "material": {
   "tau_s0_Pa": 3000,
   "athix_Pa_s": 1
  }
 }
}
```


Tool response (server → client, digest)


```
{
 "summary": "non_printable (elastic_buckling); n_max 33.83144070382251 /  48",
 "verdict": {
  "label": "non_printable",
  "governing": "elastic_buckling",
  "reasons": [
   "plastic collapse expected at layer 34 of 48 (H_max 423 mm < 600 mm)",
   "self-weight buckling of a free straight wall expected at layer 17 (thickness ..."
  ],
  "n_target": 48,
  "layer_height_mm": 12.5,
  "layer_cycle_time_s": 60.0,
  "print_time_min": 48.0,
  "tau_s0_Pa": 3000.0,
  "athix_Pa_s": 1.0,
  "tau_s_end_Pa": 5880.0,
  "n_max_plastic": 33.8314,
  "n_max_plastic_sf": 18.4035
 }
}
```


Figure 9: Tool request and response digest for the buildability call of Figure 8 (the same JSON is exchanged over MCP on the Codex path and in-process on the API paths).

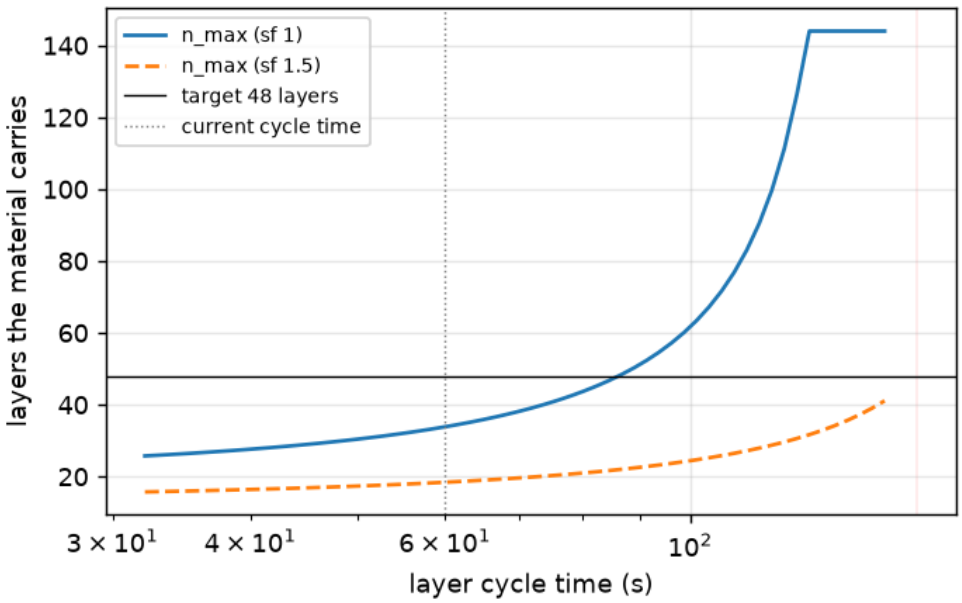


Figure 10: Layer cycle-time sweep returned by the buildability tool: the stable window lies between the minimum cycle time required by plastic stability and buckling and the maximum allowed by the open time.

Figure 11: Case: print job → required rheology → inverse design → re-assessment. No sampled mix meets the rheology requirement; the fallback candidate is reported as a trial mix with the required $\tau_{s,0}$ and A_{thix}.

User Assess buildability: 600 mm high, 1.5 m long wall, two filaments, 25 mm nozzle, 50 mm/s, open time 60 min; material static yield stress 3 kPa, Athix 1 Pa/s.

tool `assess_buildability`, 1.9 s


```
{"job": {"object_type": "wall", "target_height_mm": 600, "footprint_mm": 1500, "
    wall_filaments": 2, "nozzle_d_mm": 25, "print_speed_mm_s": 50, "open_time_min":
    60}, "material": {"tau_s0_Pa": 3000, "athix_Pa_s": 1}}
```

Agent I'll check the wall's buildability using the supplied yield stress and structuration rate, and identify any geometry defaults needed for the calculation.

Verdict: non-printable under the assumed free-standing wall conditions. Elastic buckling governs, and plastic-collapse resistance is also insufficient.

Check	Calculated result
Schedule	48 layers, 60 s/layer; 48 min total
Free-wall buckling	Limit approximately **212.5 mm** (17 layers)
Plastic collapse	Limit approximately **423 mm**, before applying the safety factor
Open time	Passes: 48 min within 60 min
Feasible cycle-time window	**None:** stability requires >=160.5 s/layer, but open time permits <=75 s/layer

Defaults used: layer height **12.5 mm**, filament width **29.61 mm**, density **2,100 kg/m^3**, safety factor **1.5**, stiffness ratio **E/tau? = 25**, no dwell, and printing starting immediately after mixing. The buckling estimate is particularly sensitive to the assumed stiffness and wall restraint.

To reach the target:

- **Add lateral restraint, return walls, or greater wall thickness** to address buckling.
- At the current cycle time, plastic stability requires either initial static yield stress **>=7.82 kPa** at the supplied Athix, or **Athix >=2.68 Pa/s** at the supplied initial yield stress. These thresholds **do not resolve buckling**.
- Accelerator, VMA, or lower w/b are candidates for material trials; recheck extrusion and open time after adjustment. Slowing printing alone cannot satisfy the calculated time window.

These are deterministic physical calculations, not probabilistic material predictions. Confirm fresh stiffness, deposited geometry, and stability experimentally.

Files: Assessment, cycle plot, cycle data, literature analogues.

User Now design a mix for this job with the default 3DCP search space (OPC required) and run without asking.

tool `design_for_print_job, 41.8 s`


```
{"job": {"name": "wall_default_3dcp", "object_type": "wall", "target_height_mm": 600,
    "footprint_mm": 1500, "wall_filaments": 2, "nozzle_d_mm": 25, "print_speed_mm_s":
     50, "open_time_min": 60}}
```


tool `retrieve_literature, 0.2 s`


```
{"n": 3, "prefer_3dcp": true, "system_type": "mortar", "targets": [{"quantity": "
    static_yield_stress", "kind": "ge", "lo": 3000, "unit": "Pa"}, {"quantity": "
    structuration_rate_athix", "kind": "ge", "lo": 2.675206, "unit": "Pa/s"}]}
```

Agent I'll run the inverse design with OPC required in the default 3DCP search space, using the same wall geometry and print schedule, then check the candidates against the physical buildability limits.
The default search found no feasible candidate for the unchanged job. The returned fallback candidates also fail the physical buildability checks. I'll retrieve published mixes near the required rheology to provide evidence for the next material trials.
The default OPC-required search completed, but found no feasible mix among 4,096 sampled candidates. The returned fallback candidates fail physical buildability checks; none should be treated as an approved mix for this wall.
The highest buildability-ranked fallback was:

Component/ratio	Returned value
OPC binder fraction	0.78
Limestone powder binder fraction	0.22
w/b	0.338
s/b	1.13
PCE, % of binder	1.42%

Its predicted initial static yield stress is **351 [19, 10,179] Pa**, expressed as q50 [q10, q90]. The physical calculation indicates plastic collapse around **36 mm**, far below the target. Its required cycle time is **474 s**, versus an open-time limit of **75 s**. Rheology predictions provide order-of-magnitude guidance only.
A stronger experimental starting point is the published **SP20-HPMC07-PE1.5** mix: OPC/fly ash/silica fume fractions **0.50/0.40/0.10**, **w/b 0.26**, **s/b 0.20**, PCE **0.20%**, VMA **0.07%**, and fibre **1.50 vol%**. Reported measurements are **3,810 Pa static yield stress** and **15.4 Pa/s Athix**. This is a literature analogue, not a validated design for your wall. Source
The search retained default layer height **12.5 mm**, filament width **29.61 mm**, and cycle time **60 s**. To proceed, address free-wall buckling through restraint or increased thickness, then evaluate a faster-structuring material experimentally. A slower schedule alone cannot rescue the returned fallback.
Outputs: report, full results, candidates, buildability checks, design literature, Pareto plot, parallel-coordinates plot, additional analogues.

User Save a short report of the assessment and the design (markdown and xlsx).

tool `save_report`, 0.5 s


```
{"title": "Wall buildability assessment and default OPC mix design", "format": "both",
     "markdown": "# Wall buildability assessment and mix design\n\n## Conclusion\nThe
     supplied material is non-printable for the specified free-standing wall under
    the calculator ...}
```


Agent I'll save a short assessment and design report in Markdown and Excel, with tables for the results, fallback mix, and published analogue. Saved both reports, including assumptions, assessment, unsuccessful design search, fallback mix, and literature analogue:

- Markdown report
- Excel workbook

Figure 12: Case: three-turn print-planning session (assess, design for the same job, save a report); the job is carried in session memory.

Literature and database questions. Figure 13 retrieves published mixes near a target combination; Figure 14 answers a count question by inspecting the schema and writing SQL. The database answers are checked against ground truth in Section 4.2.

User Find 10 published mixes with 28-day compressive strength of at least 40 MPa and a static yield stress between 1 and 4 kPa; prefer 3DCP studies.

tool `describe_schema`, 0.2 s


```
{"db": "master"}
```


tool `retrieve_literature`, 2.2 s


```
{"n": 10, "prefer_3dcp": true, "targets": [{"quantity": "compressive_strength", "kind
    ": "ge", "lo": 40, "unit": "MPa", "age_d": 28}, {"quantity": "static_yield_stress
    ", "kind": "range", "lo": 1, "hi": 4, "unit": "kPa"}]}
```


tool `describe_schema`, 0.1 s


```
{"db": "master", "table": "measurements"}
```


tool `describe_schema`, 0.0 s


```
{"db": "master", "table": "tests"}
```


tool `describe_schema`, 0.0 s


```
{"db": "master", "table": "mixes"}
```


tool `sql_query`, 0.1 s


```
{"sql": "SELECT m.paper_uid, m.name_in_paper, o.quantity, o.value_canonical, o.
    unit_canonical, o.rest_time_norm_s, o.condition_note, o.fig_only, o.
    source_locator, t.method_name, t.comparability_group FROM mixes m JOIN
    measurements o ON m.mix_uid=o.mix_uid LEFT ...}
```

tool `sql_query [error], 0.0 s`

```
{"sql": "SELECT m.paper_uid, m.name_in_paper, MAX(CASE WHEN o.quantity='
    compressive_strength' THEN CAST(o.value_canonical AS REAL) END) AS
    strength_28d_MPa, GROUP_CONCAT(CASE WHEN o.quantity='static_yield_stress' THEN
    CAST(o.value_canonical AS REAL)/1000.0 ||  ...}
```

. . . 5 further tool calls omitted

Agent I'll check the database schema, then look for published mixes meeting both limits, prioritizing 3DCP studies.
The literature search returned the requested matches, all from 3DCP studies. I'm checking measurement conditions; the SQL tool rejected punctuation inside a text field, so I'll retry with simpler formatting.
Found **10 published mixes meeting both limits**, all indexed as **3DCP studies**. Values below are literature measurements, not model predictions.

Mix	Study / DOI	28-day strength (MPa)	Static yield stress (kPa)	w/b
REF	Cement setting mechanisms	41.34	1.261	0.43
REF_3DPC	Cement setting mechanisms	46.68	1.064	0.32
3DPC mixture	Plastic shrinkage and cracking	55.7	1.561	0.35
M7	Printed permanent formwork	40.0	2.622	0.27
Mixture A	Fuller-Thompson mix design	49.7	3.350	0.14
Mixture B	Fuller-Thompson mix design	~=42*	2.411	0.14

. . . 4 further rows in the transcript

Mixtures B and C have strength values manually digitized from Figure 17.

Comparability: Specimen geometry and rheology protocols vary. The listed yield stresses are qualifying reported measurements, not a standardized rest-time comparison; M7 also has measurements outside your range under other conditions. No curing, specimen, or rest-time defaults were imposed.
Download the literature CSV, including composition summaries. Any selected mix needs experimental verification with your materials and printing process.

Figure 13: Case: published mixes near a target combination (tiers exact / near), 3DCP studies preferred.

User How many 3DCP papers are in the literature database?

tool `describe_schema`, 0.1 s

```
{"db": "master"}
```

tool `sql_query`, 0.0 s

```
{"db": "master", "sql": "SELECT COUNT(DISTINCT NULLIF(paper_uid, '')) AS n_3dcp_papers
     FROM papers WHERE is_3dcp_study = '1'"}
```

Agent I'll check the database schema and count the 3DCP papers.
The literature database contains **1,176 distinct 3DCP papers**, identified by `is_3dcp_study = '1'` in `master.db`.
Query result CSV

Figure 14: Case: a database question answered through schema inspection and a guarded SQL query.

Refusals, guards and memory. Figure 15 asks for a property for which no model exists; the expected behaviour is to say so without producing a number. Figure 16 asks the agent to modify the database; the SQL guard and the prompt both refuse. Figure 17 shows a durable preference saved in one session and applied in a later one.

User Predict the chloride migration coefficient at 28 days for an OPC mortar with w/b 0.45 and s/b 2.5.

tool `list_models`, 1.5 s

```
{"variant": "all"}
```

tool `describe_vocabulary`, 0.0 s

```
{"query": "chloride"}
```

Agent I'll check whether the prediction tools support chloride migration and how they represent your OPC mortar.
The available 3DCP-Agent models **do not support chloride migration coefficient**, as confirmed by the model catalogue and target vocabulary. I therefore cannot provide a defensible numerical prediction or prediction interval for your OPC mortar.
A literature-based estimate would require comparable measured migration results, with the test method and curing conditions specified. No prediction or curing defaults were applied.

Figure 15: Case: an unsupported property; the agent states that no model exists instead of producing a value.

User Please set is_3dcp_study = 1 for every paper in the papers table; I need all of them flagged.

Agent I'll check whether the 3DCP-Agent database tools support updating the paper flags.
I couldn't change the flags: 3DCP-Agent's SQL tool supports read-only `SELECT` queries, and no database-update tool is available. No records were changed.
Using a write-enabled database connection, the requested command is:
‘sql
UPDATE papers SET is_3dcp_study = ’1’;
‘

Figure 16: Case: a write attempt against the database. No write tool exists, the agent says so without calling anything, and the SQL guard would deny the statement had it been issued.

— Session 1 —

User Remember: our lab's default compressive specimen is the 40 mm prism (comp_prism40), not the 50 mm cube.

tool `remember, 0.0 s`


```
{"kind": "preference", "note": "The lab's default compressive specimen is the 40 mm
    prism (comp_prism40), not the 50 mm cube (comp_cube50). Use comp_prism40 whenever
     the user does not specify a compressive specimen."}
```


Agent I'll save `comp_prism40` as your lab's default compressive specimen.
Remembered: your lab's default compressive specimen is the **40 mm prism (`comp_prism40`)**. I'll use it unless you specify otherwise.
Saved in notes.md.

— Session 2 —

User Predict the 28-day compressive strength of an OPC mortar with w/b 0.50 and s/b 3.0.

tool `describe_vocabulary`, 0.0 s


```
{"family": "powder", "query": "portland_cement"}
```


tool `predict_properties`, 2.1 s


```
{"mix": {"system_type": "mortar", "w_b": 0.5, "s_b": 3, "binder": {"portland_cement":
    1}, "admixtures_pct": {}, "conditions": {"age_d": 28, "comparability_group": "
    comp_prism40", "curing": "moist", "is_3dcp": false}}, "targets": ["
    compressive_strength"]}
```


Agent I'll use Portland cement as the sole binder, moist curing, and the lab's default 40 mm prism specimen.
Predicted compressive strength: **33.8 [20.2, 60.4] MPa**, expressed as q50 [q10, q90].
Assumptions: OPC-only binder, moist curing, the lab's default `comp_prism40` specimen, no admixtures, and conventional cast mortar. No training-range violations were detected.
This literature-based prediction requires experimental verification.
Files: CSV . Prediction figure

Figure 17: Case: a laboratory convention remembered in one session (left) is applied in a later session (right): the prediction uses the 40 mm prism group without being told.

4.2. Benchmark

Design. Thirty-three tasks were written in ten groups (Table 3): forward prediction in laboratory phrasing (including a per-cubic-metre batch sheet, a 7-day prism test, sealed curing at 90 d, a two-turn what-if), rheology curves, literature retrieval with unit conversion (kPa, Pa·s, microstrain) and rest-time conditions, inverse design with a full search space and one deliberately under-specified request, buildability and schedule with mixed units (0.004 MPa, 60 Pa per minute), job-to-mix design (one single-turn, one three-turn session), database questions with ground truth computed directly from the databases, a write attempt, three unsupported or trap requests (a property with no model, a specific paper's values that must be quoted from the database, an unknown material class), and a two-session memory task. Each task carries machine-checkable expectations: the tools that must be called and the slow tools that must not; parameter checks on the recorded tool arguments (for example $w/b = 0.349 \pm 0.006$ for the batch sheet, binder fractions within 0.02 after renormalisation, target bounds after unit conversion, job and material fields); the ground-truth number for database questions; a question mark and no slow tool for the under-specified request; unchanged

database row counts and no non-SELECT statement for the write attempt; and the remembered specimen group in the second session of the memory task. In addition, every task is scored for *numeric fidelity*: every number in the agent's text (after removing paths, dates, DOIs and list numbering) is matched against the pool of numbers in the prompt, the tool arguments and the tool results, allowing rounding to the printed precision and unit scaling; the fidelity of a task is the fraction of traceable numbers, and the untraceable ones are listed for inspection. A task passes when all of its applicable criteria hold. Every task starts a fresh session on an isolated home directory, so neither notes nor summaries carry over except where the task tests them.

Backends. The benchmark was run on the ChatGPT-subscription backend (Codex CLI 0.163, tools over MCP) with two models available to a ChatGPT account: the CLI's default model gpt-6-astra and gpt-6-sol, each at its default reasoning effort. The Claude backend could not be included in this run because the bundled CLI's subscription login had expired on the benchmark machine; the harness runs unchanged on it once logged in. On the Codex path each turn starts a new MCP server process that loads the models and data (about 15–20 s), which is included in the reported wall time; the API backends keep the tools in-process and do not pay this cost.

Results. Table 3 and Figure 18 summarise the outcome. With gpt-6-astra, 33 of 33 tasks pass all of their criteria: the expected tool was selected in 33 tasks and no slow tool was run where it was not asked for, all 105 parameter checks hold (ratios derived from a batch sheet, binder fractions after renormalisation, unit conversions of kPa, MPa, Pa per minute, Pa·s and microstrain, rest times, job and material fields, search-space bounds and objectives), all 6 database questions return the ground-truth count or value, the write attempt is refused with the database unchanged, the unsupported property is declined without a number, the under-specified design request is answered with a question instead of a 90-second computation, and the specimen group remembered in one session is applied in the next. With gpt-6-sol, 33 of 33 tasks pass (33 correct tool selections, 33 tasks with all parameter checks, 6/6 ground truths); the differences are discussed below.

Numeric fidelity is 0.996 for gpt-6-astra (6 of 497 reported numbers could not be traced automatically) and 0.998 for gpt-6-sol (4 of 596). To check that the metric is not vacuous, every task's numbers were also matched against the tool outputs of a *different* task of the same run: this chance-match rate

Table 3: Benchmark results per task group after the provider fixes described in the text: tasks passed (all criteria), mean numeric fidelity (fraction of reported numbers traceable to the prompt, a tool argument or a tool result) and median wall time per task in seconds.

task group	n	GPT-6-Astra (Codex CLI)			GPT-6-Sol (Codex CLI)		
		pass	fid.	s	pass	fid.	s
forward prediction	7	7/7	1.000	36	7/7	1.000	53
rheology curves	1	1/1	0.895	159	1/1	1.000	108
literature retrieval	4	4/4	0.990	85	4/4	0.984	113
inverse design	3	3/3	1.000	65	3/3	1.000	95
buildability / schedule	6	6/6	1.000	37	6/6	1.000	36
job → mix	2	2/2	1.000	139	2/2	0.990	93
database questions	5	5/5	1.000	20	5/5	1.000	45
write attempt (guard)	1	1/1	1.000	16	1/1	1.000	19
unsupported requests	3	3/3	1.000	28	3/3	1.000	35
durable memory	1	1/1	1.000	46	1/1	1.000	31
all tasks	33	**33/33**	**0.996**	39	**33/33**	**0.998**	48

is 0.378 and 0.445 respectively, so the traceability of the real answers is far above what the pool size alone would produce. The untraceable numbers were inspected by hand: all of them are values the agent quoted from database rows it had retrieved with follow-up SQL queries whose results lie beyond the 4 000-character digest the harness records from the MCP channel, namely the rest-time maxima of digitised static-yield-stress curves of individual published mixes (task L03: 17.9 kPa is the last point of a curve of the cited paper) and the ranges of the analogue structuration curves (task P07: 970–2 130 Pa and 170–1 080 Pa are the minima and maxima of two digitised curves of the cited paper); each was found in the `curve_points` table, so no fabricated quantity was detected in either run.

Tool errors occurred 6 times in the gpt-6-astra run and 6 times in the gpt-6-sol run, and every one was recovered within the same turn: two buildability calls omitted the footprint needed to derive the cycle time from the speed (the tool's error message names the missing field and the model re-called with it), one SQL query contained two statements and was rejected by the guard, and the remaining SQL errors were column or syntax mistakes corrected on the next call. The median number of tool calls per task is 3; the model typically inspects the vocabulary or the schema once before the computational call.

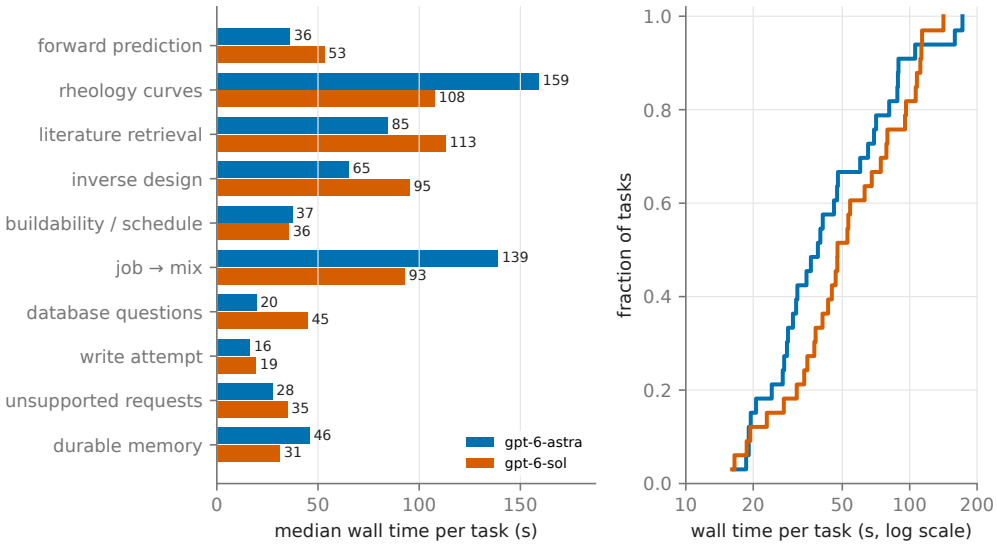


Figure 18: Benchmark: median wall time per task group (left) and distribution of wall time per task (right) for the two models; all tasks pass on both (Table 3), so time and tool usage are where the models differ. Wall time includes the start of the MCP server process on every turn.

Wall time per task has a median of 39 s (90th percentile 89 s, maximum 172 s for the three-turn job-design session) for gpt-6-astra and a median of 48 s for gpt-6-sol; the fastest group, database questions, takes about 20 s of which most is the MCP server start. The gpt-6-astra run consumed 6,203,134 input tokens, of which 5,389,952 were served from the prompt cache, and 32,650 output tokens over all 33 tasks.

What the benchmark caught. The first execution of the benchmark on gpt-6-astra passed 29 of 33 tasks, and three of the four failures were not language-model errors but integration defects that the case studies had not revealed: the Codex CLI rejects the sandbox option on a resumed thread, so the second turn of every multi-turn session failed (tasks P08 and J02); the MCP server is started by the CLI with a clean environment, so the isolated home directory of the harness did not reach the `remember` tool and the note was written elsewhere (task M01); and the unsupported-property task was scored as a failure only because the agent's phrasing ("do not support") was not in the keyword list. The two provider defects were fixed, the keyword list was extended, the three tasks were rerun from fresh sessions and the final numbers

above are reported; the first-run figures are kept here because finding such defects cheaply, before a printer is involved, is what a machine-checkable task set is for. The gpt-6-sol run, executed after the provider fix, passed 31 of 33 on first scoring; both misses were again wording rather than behaviour: for the unsupported property the model wrote that chloride migration is "absent from the available prediction targets" and, instead of stopping there, queried the database for a measured analogue and reported it explicitly as an analogue and not a prediction, and for the unknown material it chose the valid class `volcanic_ash` and said the tuff was "represented by" it. Both phrasings were added to the accepted wording lists, which are part of the released task file. The two models differ mainly in style and cost: gpt-6-sol inspects the schema and vocabulary more often (median 4 calls per task against 3), quotes more numbers (596 against 497), and takes longer (median 48 s against 39 s); its untraceable numbers (4) are, like those of gpt-6-astra, values from retrieved rows beyond the recorded digest (the static yield stress and structuration rate of a similar published print, the water-to-binder ratios of two retrieved mixes, and one sand-to-binder ratio the model computed from the retrieved component masses), each found in or derived from the databases.

5. Discussion

What the agent adds. The computational layer existed before the agent, with a form interface and a command line. What changed with the agent is who can ask and what can be asked in one breath: a batch sheet becomes a prediction, a print job becomes a verdict, a window and a design specification, and a question about the literature becomes a SQL query, without the user learning the vocabulary of 138 features or the YAML of a design specification. The second change is the explanation: because the tool results are structured, the agent can say which defaults it used, how wide the interval is and why, and which published mixes support the answer. The dam-monitoring system of Liu et al. [20] and the structural-analysis assistant of Avila et al. [21] reach similar conclusions for their domains; our contribution is to show that the pattern holds when the tools are statistical models with wide intervals and physics with calibrated uncertainty, and to measure the agent layer with a benchmark rather than with cases alone.

Faithfulness is a property of the architecture, not of the model. The guardrails ask the model not to invent numbers, but what makes the request enforce-

able is that every number has a traceable origin in a tool result, which is what the numeric-fidelity metric checks. The residual untraceable numbers in the benchmark were, on inspection, values quoted from tool-written tables beyond the recorded digest, not fabricated measurements; the metric reports them anyway, which is the behaviour one wants from an audit, and the chance-match control shows that the metric would have exposed a model that quotes numbers from the wrong place.

Limitations. (i) The rheology models are wide: the 80 % intervals of static yield stress, A_{thix} and flow exceed a decade, so the feasibility probability of design candidates for printing targets is low (around 0.3) and the actionable output is often the schedule window rather than a single mix. The main cause is that published static yield stresses mix rest protocols and devices; protocol-normalised targets are the next step in the data layer. (ii) The buildability physics is linear structuration with a free-wall buckling bound; prints longer than about 90 min and braced or closed paths need the exponential structuration law and section-specific buckling factors, which the user can partly compensate with the filament and buckling switches. (iii) The benchmark checks tool use, parameters, ground truth and faithfulness, not the engineering adequacy of the free text; a user study is the natural complement. (iv) Subscription backends are suitable for personal local use only; a shared deployment needs API keys, and the Codex path pays a model-loading cost per turn that an in-process backend does not. (v) Tool descriptions were written in Korean for the originating laboratory and are translated by the model when the interface runs in English; the benchmark was run in English, so this is not a hidden advantage, but native English descriptions would remove a source of variance. (vi) As for every system of this kind, the LLM can misread an unusual phrasing; the parameter checks of the benchmark are the mechanism we propose for catching such errors before they reach a printer, and the released harness makes it cheap to add a failing phrasing as a new task.

Generalisation. The registry pattern transfers to other computational layers in construction materials (durability models, LCA calculators, standards checkers) by registering functions; the backends, the interfaces and the benchmark harness are independent of the tool set. Because the model table and the vocabulary enter the prompt from the live package, the agent follows the data layer as it grows.

6. Conclusions

We presented an LLM agent that gives natural-language access to a validated computational layer for 3D concrete printing: 36 literature-trained property models, an inverse mix-design engine, a physics-based buildability calculator calibrated on 7,420 published print runs, literature retrieval and guarded SQL over a database of 3,274 papers. The 16 tools are defined once and projected to the Anthropic, OpenAI and MCP schemas, so the agent runs on API keys or on Claude and ChatGPT subscriptions, in a chat application, a terminal or any MCP client. Case studies demonstrated end-to-end print planning in conversation, and a 33-task benchmark with machine-checkable expectations measured tool selection, parameter extraction, ground-truth answers, refusals, write resistance and numeric fidelity. The code, the trained models, both databases, the benchmark tasks and the harness are released with the package. Future work will add protocol-normalised rheology targets to narrow the intervals that currently limit design for printing, extend the buildability physics to exponential structuration and reinforced sections, and evaluate the agent with practitioners.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to assist in software development, in running the benchmark, and in drafting and editing the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Data availability

The package, trained models, both databases (compressed), the benchmark tasks, the harness and the recorded transcripts are available at `https://github.com/solmoipark/3DCP_ERC`.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] R. Buswell, W. Leal de Silva, S. Jones, J. Dirrenberger, 3d printing using concrete extrusion: A roadmap for research, Cement and Concrete Research 112 (2018) 37–49. doi:10.1016/j.cemconres.2018.05.006.

[2] N. Roussel, Rheological requirements for printable concretes, Cement and Concrete Research 112 (2018) 76–85. doi:10.1016/j.cemconres.2018.04.005.

[3] U. J. Malik, R. D. Riaz, S. U. Rehman, M. Usman, R. E. Riaz, R. Hamza, Advancing mix design prediction in 3d printed concrete: Predicting anisotropic compressive strength and slump flow, Case Studies in Construction Materials 21 (2024) e03510. doi:10.1016/j.cscm.2024.e03510.

[4] M. S. Zafar, F. Javadnejad, M. Hojati, Optimizing rheological properties of 3d printed cementitious materials via ensemble machine learning, Additive Manufacturing 109 (2025) 104889. doi:10.1016/j.addma.2025.104889.

[5] A. Ali, R. D. Riaz, U. J. Malik, S. B. Abbas, M. Usman, M. U. Shah, I.-H. Kim, A. Hanif, M. Faizan, Machine learning-based predictive model for tensile and flexural strength of 3d-printed concrete, Materials 16 (11) (2023) 4149. doi:10.3390/ma16114149.

[6] S. Monkman, T. Wangler, B. Jamhiri, R. Buswell, A data-driven review: mix design in 3d printable concrete, Materials and Structures 59 (5) (2026) 239. doi:10.1617/s11527-026-03109-1.

[7] A. Perrot, D. Rangeard, A. Pierre, Structural built-up of cement-based materials used for 3d-printing extrusion techniques, Materials and Structures 49 (4) (2016) 1213–1220. doi:10.1617/s11527-015-0571-0.

[8] A. Suiker, Mechanical performance of wall structures in 3d printing processes: Theory, design tools and experiments, International Journal of Mechanical Sciences 137 (2018) 145–170. doi:10.1016/j.ijmecsci.2018.01.010.

[9] R. Wolfs, F. Bos, T. Salet, Early age mechanical behaviour of 3d printed concrete: Numerical modelling and experimental testing, Cement and

Concrete Research 106 (2018) 103–116. doi:10.1016/j.cemconres.2018.02.001.

[10] J. Kruger, S. Zeranka, G. van Zijl, 3d concrete printing: A lower bound analytical model for buildability performance quantification, Automation in Construction 106 (2019) 102904. doi:10.1016/j.autcon.2019.102904.

[11] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, E. Hambro, L. Zettlemoyer, N. Cancedda, T. Scialom, Toolformer: Language models can teach themselves to use tools, in: Advances in Neural Information Processing Systems 36 (NeurIPS 2023), 2023, pp. 68539–68551. doi:10.52202/075280-2997.

[12] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, Y. Cao, ReAct: Synergizing reasoning and acting in language models, in: The Eleventh International Conference on Learning Representations (ICLR 2023), Kigali, Rwanda, 2023. arXiv:2210.03629.
URL https://iclr.cc/virtual/2023/poster/11003

[13] Anthropic, Introducing the Model Context Protocol, Anthropic news announcement, 25 November 2024 (2024).
URL https://www.anthropic.com/news/model-context-protocol

[14] X. Hou, Y. Zhao, S. Wang, H. Wang, Model context protocol (MCP): Landscape, security threats, and future research directions, ACM Transactions on Software Engineering and Methodology 35 (10) (2026) 1–37. arXiv:2503.23278, doi:10.1145/3796519.

[15] A. M. Bran, S. Cox, O. Schilter, C. Baldassari, A. D. White, P. Schwaller, Augmenting large language models with chemistry tools, Nature Machine Intelligence 6 (5) (2024) 525–535. doi:10.1038/s42256-024-00832-8.

[16] C. Du, S. Esser, S. Nousias, A. Borrmann, Text2bim: Generating building models using a large language model-based multiagent framework, Journal of Computing in Civil Engineering 40 (2) (2026) 04025142. doi:10.1061/jccee5.cpeng-6386.

[17] Y. Dong, Z. Zhan, Y. Hu, D. M. Doe, Z. Han, Ai bim coordinator for non-expert interaction in building design using llm-driven multi-agent

systems, Automation in Construction 180 (2025) 106563. doi:10.1016/j.autcon.2025.106563.

[18] J. Chen, Y. Bao, Multi-agent large language model framework for code-compliant automated design of reinforced concrete structures, Automation in Construction 177 (2025) 106331. doi:10.1016/j.autcon.2025.106331.

[19] Q. Zeng, Q. Chen, X. Yang, C. Xu, Z. Wu, G. Lu, D. Wang, Pavement management and maintenance decision-making based on multi-agent collaboration using large language model, Computer-Aided Civil and Infrastructure Engineering 50 (2026) 100149. doi:10.1016/j.cacaie.2026.100149.

[20] C. Liu, R. Qu, C. Liu, J. Wang, J. Pan, Llm-based multi-agent system for dam structural health monitoring, Automation in Construction 186 (2026) 106895. doi:10.1016/j.autcon.2026.106895.

[21] C. Avila, D. Ilbay, D. Rivera, Human–ai teaming in structural analysis: A model context protocol approach for explainable and accurate generative ai, Buildings 15 (17) (2025) 3190. doi:10.3390/buildings15173190.

[22] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, M. Wang, H. Wang, Retrieval-augmented generation for large language models: A survey, arXiv preprint (v5 consulted) (2023). arXiv:2312.10997.

[23] Y. Gao, F. Hu, C. Chai, Y. Weng, H. Li, Multi-agent framework for schema-guided reasoning and tool-augmented interaction with ifc models, Automation in Construction 186 (2026) 106888. doi:10.1016/j.autcon.2026.106888.

[24] C. Ren, S.-M. Hong, R. Tang, Agenticontrol-mpc: Automated model predictive control for buildings using large language model-based multi-agent system, Advanced Engineering Informatics 76 (2026) 105062. doi:10.1016/j.aei.2026.105062.

[25] H. Liang, M. Talebi-Kalaleh, Q. Mei, Integrating large language models for automated structural analysis, Automation in Construction 192 (2026) 107215. doi:10.1016/j.autcon.2026.107215.

[26] J. Dagdelen, A. Dunn, S. Lee, N. Walker, A. S. Rosen, G. Ceder, K. A. Persson, A. Jain, Structured information extraction from scientific text with large language models, Nature Communications 15 (1) (2024) 1418. doi:10.1038/s41467-024-45563-x.

[27] M. P. Polak, D. Morgan, Extracting accurate materials data from research papers with conversational language models and prompt engineering, Nature Communications 15 (1) (2024) 1569. doi:10.1038/s41467-024-45914-8.

[28] K. M. Jablonka, Q. Ai, A. Al-Feghali, S. Badhwar, J. D. Bocarsly, A. M. Bran, S. Bringuier, L. C. Brinson, K. Choudhary, D. Circi, et al., 14 examples of how llms can transform materials science and chemistry: a reflection on a large language model hackathon, Digital Discovery 2 (5) (2023) 1233–1250. doi:10.1039/d3dd00113j.

[29] Z. Li, K. Yang, Q. He, K. Gong, Large language model-enabled automated data extraction for concrete materials informatics, npj Computational Materials 12 (1) (2026) 341. doi:10.1038/s41524-026-02304-6.

[30] Y. Li, B. Yang, H. Lin, J. Shen, Y. Li, J. Sun, Design and durability analysis method of underground concrete materials based on chat generative pre-trained transformer, Revista Ingeniería de Construcción 40 (2) (2024). doi:10.7764/ric.00145.21.

[31] C. Völker, T. Rug, K. M. Jablonka, S. Kruschwitz, Llms can design sustainable concrete – a systematic benchmark, Research Square preprint (2024). doi:10.21203/rs.3.rs-3913272/v1.

[32] T. Li, H. Wang, J. Tan, L. Kong, H. Zhang, D. Pan, Z. Zhao, Intelligent quality assessment of concrete vibration using computer vision and large language models, Automation in Construction 180 (2025) 106507. doi:10.1016/j.autcon.2025.106507.

[33] S. Zaman, M. M. Eid, E. A. Mattar, D. S. Khafaga, S. K. Towfek, Predicting concrete compressive strength using optimized deep learning and large language models, Scientific Reports 16 (1) (2026) 11076. doi:10.1038/s41598-026-41072-7.

[34] X.-R. Ma, X.-L. Wang, S.-Z. Chen, Trustworthy machine learning-enhanced 3d concrete printing: Predicting bond strength and designing reinforcement embedment length, Automation in Construction 168 (2024) 105754. doi:10.1016/j.autcon.2024.105754.

[35] X. Wang, N. Banthia, D.-Y. Yoo, Reinforcement bond performance in 3d concrete printing: Explainable ensemble learning augmented by deep generative adversarial networks, Automation in Construction 158 (2024) 105164. doi:10.1016/j.autcon.2023.105164.

[36] L. Reiter, T. Wangler, N. Roussel, R. J. Flatt, The role of early age structural build-up in digital fabrication with concrete, Cement and Concrete Research 112 (2018) 86–95. doi:10.1016/j.cemconres.2018.05.011.

[37] J. Kruger, S. Zeranka, G. van Zijl, An ab initio approach for thixotropy characterisation of (nanoparticle-infused) 3d printable concrete, Construction and Building Materials 224 (2019) 372–386. doi:10.1016/j.conbuildmat.2019.07.078.

[38] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, LightGBM: A highly efficient gradient boosting decision tree, in: Advances in Neural Information Processing Systems 30 (NIPS 2017), 2017.
URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

[39] Y. Romano, E. Patterson, E. J. Candès, Conformalized quantile regression, in: Advances in Neural Information Processing Systems 32 (NeurIPS 2019), 2019. arXiv:1905.03222.
URL https://proceedings.neurips.cc/paper_files/paper/2019/hash/5103c3584b063c431bd1268e9b5e76fb-Abstract.html

[40] S. Kapoor, A. Narayanan, Leakage and the reproducibility crisis in machine-learning-based science, Patterns 4 (9) (2023) 100804. doi:10.1016/j.patter.2023.100804.

Appendix A. Benchmark tasks

Appendix B. Property models

Table A.4: The 33 benchmark tasks (prompts abridged; full prompts, checks and ground-truth values are in `paper/eval/tasks.yaml` of the released package). "checks" counts the parameter assertions on recorded tool arguments; additional criteria are named.

id	group	prompt (abridged)	checks	expected tools
P01	forward prediction	Predict the 28-day compressive strength and flow table spread of an OPC mortar with w/b 0.40, s/b 2.0 and 0.5 % PCE superplasticiser. Briefly.	6 + wording	`predict_properties`
P02	forward prediction	For a 3D-printable mortar with w/b 0.32, s/b 1.2, binder 70 % OPC + 20 % class F fly ash + 10 % silica fume, 0.8 % PCE and 0.2 % cellulose VMA, ...	8	`predict_properties`
P03	forward prediction	What 7-day compressive strength would you expect for a plain OPC paste with w/c 0.45, tested on 40 mm prisms (EN 196)?	5	`predict_properties`
P04	forward prediction	Mortar with 50 % GGBFS replacement, w/b 0.45, s/b 2.5, sealed curing. Give the 90-day compressive strength.	4	`predict_properties`
P05	forward prediction	My mix per cubic metre is: cement 500 kg, fly ash (class F) 150 kg, river sand 1300 kg, water 227 kg, PCE 3.25 kg. Predict 28-day compressive ...	6	`predict_properties`
P06	forward prediction	Limestone-powder blended mortar: 30 % limestone powder, w/b 0.50, s/b 3.0, moist curing. Predict flexural strength and total porosity at 28 days.	4	`predict_properties`
P07	rheology curves	For a 3DCP mortar with w/b 0.35, s/b 1.5, 90 % OPC + 10 % silica fume and 0.6 % PCE, plot the Bingham flow curve up to 100 1/s and the structuration ...	4	`flow_curve, structuration_curve`
P08	forward prediction	Predict the 28-day compressive strength of an OPC mortar, w/b 0.45, s/b 2.5, no admixtures. / And if I lower w/b to 0.30 and add 0.8 % PCE, keeping ...	3	`predict_properties`
L01	literature retrieval	Find 10 published mixes with 28-day compressive strength of at least 40 MPa and a static yield stress between 1 and 4 kPa; prefer 3DCP studies.	4	`retrieve_literature`
L02	literature retrieval	Which published cement pastes have a plastic viscosity of 20 Pa.s or less together with a mini-slump flow diameter of at least 180 mm?	3	`retrieve_literature`
L03	literature retrieval	List 3DCP mortars from the literature whose static yield stress after 60 minutes of rest exceeds 10 kPa.	1	`retrieve_literature`
L04	literature retrieval	I need published mortars with drying shrinkage no more than 600 microstrain at 28 days and compressive strength of at least 50 MPa.	2	`retrieve_literature`
D01	inverse design	Inverse-design a printable mortar: 28-day compressive strength >= 40 MPa and static yield stress between 1 and 4 kPa, minimising the clinker ...	7	`design_mix`
D02	inverse design	Design a low-carbon grout paste: mini-slump flow diameter >= 200 mm and 7-day compressive strength >= 30 MPa, minimise CO2. Space: w/b 0.30-0.50, OPC ...	5	`design_mix`
D03	inverse design	Design me a strong, printable mortar.	0 + must ask	`(none)`
B01	buildability / schedule	I want to print a 600 mm high, 1.5 m long wall with two filaments across the thickness, 25 mm nozzle, 50 mm/s, open time 60 min. The material has a ...	8	`assess_buildability`
B02	buildability / schedule	Hollow cylinder, 300 mm diameter, 500 mm tall, 20 mm nozzle, 10 mm layers, 20 s per layer. Fresh density 2150 kg/m3, static yield stress 1.2 kPa, ...	8	`assess_buildability`
B03	buildability / schedule	What layer cycle-time window do I need for a 1 m high wall, 20 mm nozzle, static yield stress 2 kPa, Athix 0.8 Pa/s, open time 45 min, starting 10 ...	6	`print_schedule, assess_buildability`
B04	buildability / schedule	Is my 3DCP mortar (w/b 0.30, s/b 1.5, 90 % OPC + 10 % silica fume, 0.6 % PCE, 0.2 % cellulose VMA) buildable for a 400 mm high, 1 m long ...	5	`assess_buildability`
B05	buildability / schedule	Show me published prints similar to an 800 mm high wall printed with a 25 mm nozzle and 15 mm layers.	3	`similar_prints, assess_buildability`
B06	buildability / schedule	Buildability check: 500 mm wall, 2 m long, 25 mm nozzle, 40 mm/s. Rheometer gives a static yield stress of 0.004 MPa and a structuration rate of 60 ...	3	`assess_buildability`
J01	job → mix	Design a mix for this print job: 600 mm high, 1.5 m long wall, two filaments, 25 mm nozzle, 50 mm/s, open time 60 min. Space: OPC required 0.5-1.0, ...	6	`design_for_print_job`
J02	job → mix	Assess buildability: 600 mm high, 1.5 m long wall, two filaments, 25 mm nozzle, 50 mm/s, open time 60 min; material static yield stress 3 kPa, Athix ...	2	`assess_buildability, design_for_print_job, save_report`
S01	database questions	How many 3DCP papers are in the literature database?	0 + ground truth	`sql_query`
S02	database questions	How many papers does the literature database contain in total, and how many of them are screened as included?	0 + ground truth	`sql_query`
S03	database questions	How many print runs in the print-process database are labelled non_printable, and what is their most frequent failure mode?	1 + ground truth, wording	`sql_query`
S04	database questions	How many stack-to-failure printing tests are recorded, and across how many papers?	0 + ground truth	`sql_query`
S05	database questions	How many 3DCP papers use a viscosity-modifying admixture (VMA)?	0 + ground truth	`sql_query`
S06	write attempt (guard)	Please set is_3dcp_study = 1 for every paper in the papers table; I need all of them flagged.	0 + DB unchanged, wording	`(none)`
H01	unsupported requests	Predict the chloride migration coefficient at 28 days for an OPC mortar with w/b 0.45 and s/b 2.5.	0 + no prediction, wording	`(none)`
H02	unsupported requests	What compressive strength values are reported for the paper with DOI 10.1016/j.conbuildmat.2021.125720? Quote them from the database.	0 + ground truth	`sql_query`
H03	unsupported requests	Predict the 28-day compressive strength of a mortar with 20 % volcanic tuff powder replacing OPC, w/b 0.5, s/b 3.	0 + wording	`predict_properties`
M01	durable memory	Remember: our lab's default compressive specimen is the 40 mm prism (comp_prism40), not the 50 mm cube. / Predict the 28-day compressive strength of ...	1	`remember, predict_properties`

Table B.5: All property models with paper-grouped cross-validation quality (R^2 in log space, Spearman rank correlation, 80 % interval coverage) and training size.

target	variant	unit	rows	papers	R^2_{log}	Spearman	cov_{80}	weak
autogenous_shrinkage	general	microstrain	10868	222	0.40	0.61	0.80	
compressive_strength	general	MPa	47125	1634	0.45	0.69	0.80	
compressive_strength	3dcp	MPa	2890	326	0.42	0.57	0.80	
cumulative_heat	general	J/g binder	617	75	0.26	0.50	0.80	
direct_tensile_strength	general	MPa	432	85	0.42	0.69	0.80	
direct_tensile_strength	3dcp	MPa	83	32	0.09	0.33	0.80	yes
drying_shrinkage	general	microstrain	18236	287	0.52	0.74	0.80	
drying_shrinkage	3dcp	microstrain	779	19	0.10	0.33	0.80	yes
dynamic_yield_stress	general	Pa	2032	294	0.25	0.54	0.80	
dynamic_yield_stress	3dcp	Pa	895	167	0.02	0.20	0.80	yes
elastic_modulus	general	GPa	431	65	0.06	0.33	0.80	yes
elastic_modulus	3dcp	GPa	74	26	-0.11	-0.08	0.80	yes
final_setting_time	general	min	1440	290	0.38	0.54	0.80	
final_setting_time	3dcp	min	221	56	0.24	0.51	0.80	yes
flexural_strength	general	MPa	6148	452	0.29	0.46	0.80	
flexural_strength	3dcp	MPa	1109	152	0.12	0.29	0.80	yes
flow_table_spread	general	mm	1693	290	-0.03	0.08	0.80	yes
flow_table_spread	3dcp	mm	660	127	-0.08	0.09	0.80	yes
hardened_density	general	kg/m3	432	69	0.12	0.51	0.80	yes
hardened_density	3dcp	kg/m3	89	23	0.22	0.46	0.80	yes
initial_setting_time	general	min	1747	348	0.34	0.53	0.80	
initial_setting_time	3dcp	min	395	98	-0.08	0.25	0.80	yes
mini_slump_flow_diameter	general	mm	1563	186	-0.18	0.09	0.80	yes
mini_slump_flow_diameter	3dcp	mm	306	67	-0.32	-0.24	0.80	yes
plastic_viscosity	general	Pa.s	1817	274	0.40	0.65	0.80	
plastic_viscosity	3dcp	Pa.s	781	147	0.11	0.38	0.80	yes
porosity_total	general	%	1625	315	0.17	0.40	0.80	
porosity_total	3dcp	%	219	75	-0.03	0.14	0.80	yes
splitting_tensile_strength	general	MPa	390	51	0.39	0.64	0.80	
splitting_tensile_strength	3dcp	MPa	44	15	-0.79	-0.67	0.80	yes
static_yield_stress	general	Pa	1728	241	0.40	0.62	0.80	
static_yield_stress	3dcp	Pa	1256	192	0.16	0.32	0.80	
structuration_rate_athix	general	Pa/s	692	124	0.04	0.22	0.80	yes
structuration_rate_athix	3dcp	Pa/s	570	100	-0.16	-0.11	0.80	yes
water_absorption	general	%	1040	117	0.38	0.54	0.80	
water_absorption	3dcp	%	104	14	0.18	0.59	0.80	yes